Assignment

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Assignment

Faculty:	Information Technology
Module Code:	ITOPA3-33
Module Name:	Operating Systems
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Total Marks:	100

Submission Week:	Week 6

This module is presented on NQF level 7.

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual assignment

This assignment contributes 40% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

Instructions to Students

1. Please ensure that you use the Assignment Template when preparing your submission (where applicable).
2. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
3. Remember to keep a copy of all submitted assignments.
4. All work must be typed.
5. Please note that you will be evaluated on your writing skills in all your assignments.
6. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
7. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

Section A

Learning Objective

Evaluate an OS with relation to management systems based on policies and algorithms.

Develop an understanding to model business and other non-software systems.

Investigate alternatives designs of OS.

Analyze and validate the architectural design software.

Assignment Topic:

Operating Systems

Scope

Week 1 - 6

Technical Aspects

NA

These restrictions apply only where the activity forms part of the published assessment criteria. Assessment-specific instructions take precedence where they expressly permit or provide a tool, dataset, configuration, model, code component or other output. Please refer to the below and apply as appropriate.

See Generally Prohibited Activities

4.1. Discipline-specific Prohibited Activities

Please ensure that you consult the prohibited activity list that is specific to this module's discipline from the four discipline options below.

4.1.1. Software Engineering

4.1.2. Network Security Engineering

4.1.3. Data Science

4.1.4. Robotics

Question 1 — 20 Marks

Study the scenario and complete the question(s) that follow(s):

The Reserve National Bank operates a large online banking system that processes thousands of customer transactions every day. Customer account information, transaction histories, loan records, and audit logs are stored on the bank's secure servers.

When a customer logs into online banking and requests a bank statement, the banking application must perform several tasks such as locate the customer's statement file stored on the hard drive, manage the file, display the information to the customer and close the file after the operation is completed.

Under normal circumstances, these operations are performed through basic Operating System Calls. These system calls allow the operating system to manage access to files, printers, memory, and other hardware resources securely.

However, a software developer proposes modifying the banking application so that it bypasses the operating system entirely and directly accesses the hard drive and printer hardware without using operating system system calls.

The developer argues that this approach may improve performance because the application would communicate directly with the hardware devices. The bank's Chief Information Officer (CIO) is concerned that bypassing the operating system could introduce serious security, reliability, and operational risks.

Source: Nawej, MC (2026)

1.1 Discuss the basic Operating System Calls used for file management and outline their importance when a customer requests a bank statement.

(8 Marks)

1.2 Critically analyse the consequences of allowing the banking application to directly access the hard drive and printer without using operating system system calls.

(6 Marks)

1.3.1 As the Chief Information Officer (CIO), would you approve the developer's proposal?

(2 mark)

1.3.2 Justify your decision with reference to the role of operating system system calls in modern computing environments.

(4 marks)

End of Question 1

Question 2 — 25 Marks

Study the scenario and complete the question(s) that follow(s):

A rapidly growing e-commerce company, ShopFast Ltd, operates a web server that handles thousands of customer requests per second during peak shopping hours. The server currently runs as a single-threaded application, meaning each incoming HTTP request (browsing products, adding to cart, processing payments) is handled one at a time. During a recent Black Friday sale, the server became unresponsive — customers experienced timeouts, abandoned carts, and the company lost significant revenue.

The engineering team has been tasked with redesigning the server as a multithreaded web server. Each incoming client request must be handled by a separate thread, allowing the server to serve multiple customers simultaneously. The server runs on a quad-core (4-core) machine.

The development team must also decide whether the new architecture exhibits task parallelism or data parallelism, and justify their choice before implementation begins.

Source: (Nawej, 2026)

2.1a) Explain why the single-threaded web server performed poorly under peak load.

(3 marks)

2.1b) In your answer, discuss the concept of blocking and how it prevents the server from serving multiple clients.

(3 marks)

2.2a) Describe THREE specific benefits that a multithreaded solution would provide to ShopFast Ltd's web server compared to the single-threaded design.

(3 marks)

2.2b) For each benefit, relate it directly to the e-commerce scenario described above.

(3 marks)

2.3a) Distinguish between task parallelism and data parallelism.

(3 Marks)

2.3b) Based on the ShopFast Ltd scenario, state which type the multithreaded web server exhibits and justify your answer with at least TWO reasons.

(3 Marks)

2.4a) Explain what a thread pool is.

(2 marks)

2.4b) Why is it preferred over per-request thread creation in a high-traffic web server

(2 marks)

2.4c) Describe the role of the task queue within the thread pool architecture.

(3 marks)

End of Question 2

Question 3 — 20 Marks

Study the scenario and complete the question(s) that follow(s):

A manufacturing company uses an automated production system consisting of four processes:

- P0 – Material Loading
- P1 – Quality Inspection
- P2 – Product Assembly
- P3 – Packaging

The processes compete for six resources:

- A = Conveyor Belt

- B = Robotic Arm
- C = Vision Sensor
- D = Packaging Machine
- E = Assembly Station (2 instances)
- F = Inspection Scanner

The following code is executed continuously.

void P0()	void P1()	void P2()	void P3()

{
while(true){
get(A);
get(B);
get(F);
 	
// critical region
// use A,B,F
 	
release(A);
release(B);
release(F);
}
}
	{
	while(true){
	get(C);
	get(E);
	get(D);
	
	// critical region
	// use C,E,D
	
	release(C);
	release(E);
	release(D);
	}
	}
		{
		while(true){
		get(F);
		get(A);
		
		// critical region
		// use F,A
		
		release(F);
		release(A);
		}
		}
			{
			while(true){
			get(D);
			get(C);
			
			// critical region
			// use D,C
			
			release(D);
			release(C);
			}
			}

Resource E has 2 instances while all other resources have 1 instance

3.1 Suppose the processes are executed line-by-line from P0 to P3. Using a Resource Allocation Graph, determine whether a deadlock can occur.

(10 Marks)

3.2 Modify the order of resource requests to eliminate the possibility of deadlock.

(5 Marks)

3.3 Explain whether deadlock prevention or deadlock avoidance would be more suitable for this manufacturing system.

(5 Marks)

End of Question 3

Question 4 35 Marks

Study the scenario and complete the question(s) that follow(s):

File Management

A company stores its employee records on a computer system. The system uses a hierarchical file system structure as shown below:

```
Root (C:)
|
├── HR
|   ├── Employees.xlsx
|   ├── Salaries.xlsx
|   └── Contracts.docx
|
├── Finance
|   ├── Budget2026.xlsx
|   └── Expenses.xlsx
|
├── IT
|   ├── Users.db
|   ├── Backup
|   |   ├── JanBackup.zip
|   |   └── FebBackup.zip
|   └── Software
|       └── Antivirus.exe
|
└── Public
    ├── Policies.pdf
    └── Forms.docx
```

4.1 Give three advantages of organizing files into directories.

(3 Marks)

4.2 Write the full path to **Antivirus.exe** and write a command prompt line that can be used to run the **Antivirus.exe** by considering the full path.

(4 Marks)

4.3 An administrator performs the following actions:

- Copies Policies.pdf to the HR directory.
- Deletes JanBackup.zip.
- Renames Expenses.xlsx to Expenses_2026.xlsx.

What will be the content of the HR and Public directory?

(4 Marks)

4.4 Some systems automatically delete all user files when a user logs off or a job terminates, unless the user explicitly requests that they be kept. Other systems keep all files unless the user explicitly deletes them. Discuss the relative merits of each approach.

(8 Marks)

4.5.a Why do some systems keep track of the type of a file, while still others leave it to the user and others simply do not implement multiple file types? (4 Marks)

b. Which system is "better"?

(4 Marks)

4.6a How do caches help improve performance?

(4 Marks)

b. Why do systems not use more or larger caches if they are so useful?

(4 Marks)

End of Question 4

How AI use Affects Marking

Finding	Outcome
No AI used and accurately declared	Your lecturer will mark according to the marking guide.
Permitted AI use accurately declared	Your lecturer will mark according to the marking guide.
AI-generated in identifiable assessed component	Your lecturer will mark authentic work and will not award marks for the affected component that is identified to be AI-generated.
AI-generated in dependent criteria may be affected	Your lecturer will evaluate each criterion in isolation and will not award marks for affected criteria that are AI-generated.
Technically correct work that raises authenticity concerns*	Your lecturer will complete the review before finalising the mark, and will not award marks for affected criteria that are AI-generated.
Serious authenticity concerns exist*	Your lecturer may request a formal discussion with you, and depending on the outcome may escalate a case for formal academic-integrity hearing.

*See General Prohibited Activity list under Terms of AI use for this Assessment